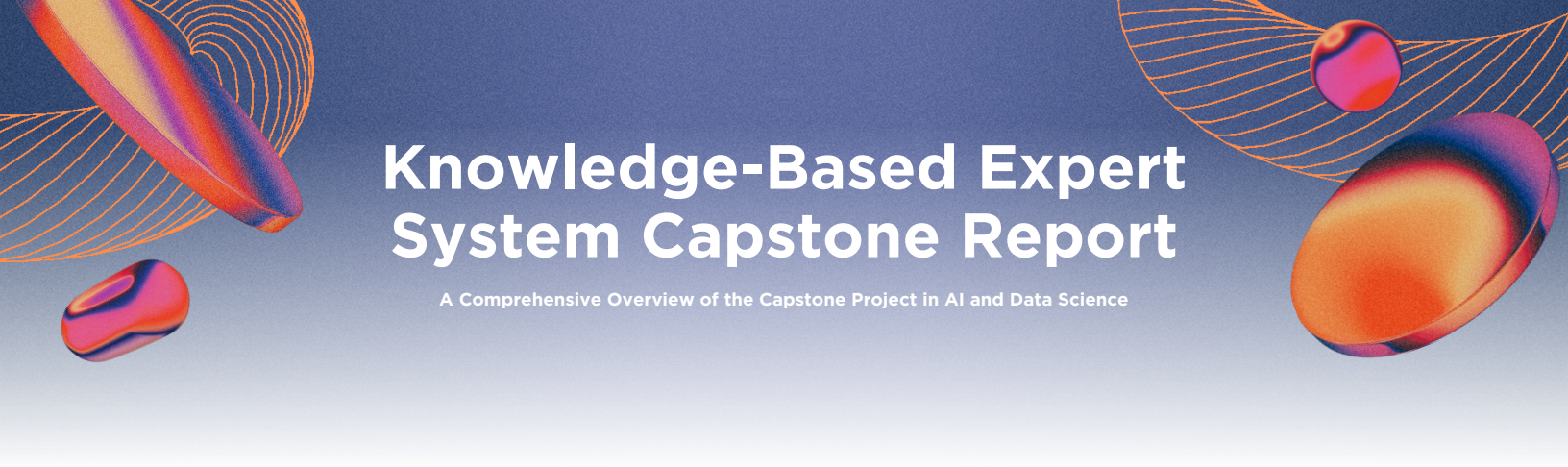




Knowledge-Based Expert System Capstone Report

A Comprehensive Overview of the Capstone Project in AI and Data Science

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1. Introduction

A Knowledge-Based Expert System is an Artificial Intelligence system designed to mimic human decision-making by leveraging stored knowledge and logical rules. This capstone project showcases a rule-based AI system that aids students in selecting the most appropriate academic course projects based on their individual interests, skills, and abilities. Unlike systems that employ machine learning, this expert system utilizes symbolic reasoning via IF-THEN rules.

2. Domain Definition

Course Project Recommendation System

Selecting a suitable project can be challenging for students due to the multitude of available choices, such as AI, web development, data science, and management systems. The system offers project recommendations based on the following criteria:

- **Programming skills**
- **Interest in AI**
- **Web development skills**
- **Creativity**
- **Problem-solving ability**
- **Communication skills**
- **Teamwork preference**

3. Problem Statement

Students frequently choose projects without adequately analyzing whether those projects align with their skills and interests, leading to:

- Poor project performance
- Lack of interest during development
- Difficulty in completing the project

Objective: Design a rule-based expert system that recommends suitable course projects based on user inputs.

4. Knowledge Representation

The system employs symbolic AI representation, consisting of:

4.1 Facts

User-provided inputs such as:

- LikesProgramming(Student)
- InterestedInAI(Student)

4.2 Predicates

- SuitableForChatbotProject(Student)
- SuitableForPortfolioWebsite(Student)

4.3 Recommendation Output

- Recommend(Project, Student)

5. Knowledge Base (IF-THEN Rules)

Rule 1:

IF LikesProgramming AND InterestedInAI
THEN SuitableForChatbotProject

Rule 2:

IF GoodProblemSolving AND InterestedInAutomation
THEN SuitableForExpertSystem

Rule 3:

IF InterestedInWebDevelopment AND Creative
THEN SuitableForPortfolioWebsite

Rule 4:

IF InterestedInDataAnalysis AND LikesProgramming
THEN SuitableForDashboardProject

Rule 5:

IF InterestedInWebDevelopment AND LikesDesign
THEN SuitableForResumeBuilder

Rule 6:

IF PrefersTeamWork AND GoodCommunication
THEN SuitableForManagementSystem

Rule 7:

IF InterestedInAI AND GoodProblemSolving
THEN SuitableForExpertSystem

Rule 8:

IF SuitableForChatbotProject
THEN Recommend(AIChatbotProject)

Rule 9:

IF SuitableForExpertSystem
THEN Recommend(KnowledgeBasedExpertSystem)

Rule 10:

IF SuitableForPortfolioWebsite
THEN Recommend(PortfolioWebsiteProject)

Rule 11:

IF SuitableForDashboardProject
THEN Recommend(DataVisualizationDashboard)

Rule 12:

IF SuitableForResumeBuilder
THEN Recommend(ResumeBuilderWebsite)

Rule 13:

IF SuitableForManagementSystem
THEN Recommend(StudentManagementSystem)

6. Inference Mechanism (Forward Chaining)

This system employs Forward Chaining, a data-driven reasoning approach that progresses from known facts to conclusions.

Process:

- Start with user facts
- Match facts with IF conditions
- Fire applicable rules
- Generate new facts
- Continue until a recommendation is produced

Why Forward Chaining?

- Works from input to conclusion
- Easy to implement in rule-based systems
- Suitable for decision-making applications

7. Sample Consultation (Step-by-Step Trace)

Student: Riya

Input Facts:

- LikesProgramming(Riya)
- InterestedInAI(Riya)
- GoodProblemSolving(Riya)
- InterestedInAutomation(Riya)
- InterestedInWebDevelopment(Riya)

Step 1:

Rule 1 fires → Chatbot Project identified

Step 2:

Rule 7 fires → Expert System identified

Step 3:

Rule 8 fires → Recommendation generated

8. System Output

Final Recommendations:

- ✓ AI Chatbot Project
- ✓ Knowledge-Based Expert System

9. Conclusion

This project successfully demonstrates a Knowledge-Based Expert System using rule-based reasoning. It illustrates how AI can assist students in decision-making without the need for machine learning. The system effectively utilizes:

- Knowledge representation
- IF-THEN rules
- Forward chaining inference

10. Limitations

- Cannot handle uncertainty
- Limited number of rules
- No learning capability
- No natural language processing
- Manual knowledge updates required

11. Future Scope

- Integration with machine learning
- Chatbot-based interface
- Database connectivity
- Personalized recommendation engine
- Web-based AI assistant

12. References

- *Artificial Intelligence: A Modern Approach* – Russell & Norvig
- *Expert Systems Principles and Programming*
- Course lecture notes
- Online AI documentation resources